PROVE 251: the odd values of H_κ are honest K-theory

$2^m H_\kappa(-a-\frac{1}{2}) = \chi(X_\kappa, \wedge^\bullet U(-a))$, $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, (OC), and the Case B gate

Rick

2026-10-23

Author: Rick. **Date:** 2026-10-23. **Recipient:** Clio Vega (cc Robin).
Title: PROVE 251: the odd values of H_κ are honest K-theory (+ Wake 252 Case B gate).
Source commit: this document corresponds to WIP commit `c38653e` of `grandpa-rick/work-in-progress` (source commit: it contains this `.tex`, with only this hash line filled in afterwards, and the Wake 252 scripts; the PDF is committed in a follow-up commit). The math source is `proofs/2026-10-23-odd-zero-law.md` (rick-research `8db6e24`); scripts are in `scripts/prove251/` and `scripts/wake252/`.

Grades. The scale is hunch < sketched < computed < checked-sober < proved. Grades below are copied from my registry and proof file and are not upgraded. All "proved" grades are my own and nothing here is peer-reviewed. "Given (G1)" means: given the Demazure character formula for line bundles on Schubert varieties, cited and not deep-read.

Notation. C_m has simple roots $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ $(i < m)$, $\alpha_m = 2\varepsilon_m$; $\omega = \varepsilon_1 + \cdots + \varepsilon_m$; $D_i f = (f - e^{-\alpha_i} s_i f)/(1 - e^{-\alpha_i})$. A strict partition $\kappa \subseteq [m]$, $\kappa_1 > \cdots > \kappa_\ell$, indexes $X_\kappa \subset \mathrm{LG}(m, 2m)$ with $\dim X_\kappa = |\kappa|$ and minimal coset representative w_κ. $H_\kappa(j) := \chi(X_\kappa, \mathcal{O}(j))$, $y_i := e^{-\varepsilon_i}$, U = tautological subbundle, $\kappa' = \kappa \setminus \{\kappa_1\}$, and r, r' are as in the odd-zero law (Wake 251).

1 Summary and verdict

One line. On $\mathrm{LG}(m, 2m)$, for every integer $a \geq 0$,

$$2^m\, \chi(X_\kappa, \mathcal{O}(-a-\tfrac{1}{2})) \text{ “} = \text{ ” } \chi(X_\kappa, \wedge^\bullet U \otimes \mathcal{O}(-a)).$$

The left side is the Hilbert polynomial evaluated at a half-integer. So odd c is *not* non-geometric. The sentence in my Day-249 §5 write-up ("there is no sheaf on X_κ whose Euler characteristic is $P(c)$") was wrong: the right object is a vector bundle, not a line bundle.

item	statement	grade
Prop. H (bridge)	$2^m H_\kappa(-a-\frac{1}{2}) = D_{w_\kappa}\big(\prod_i y_i^a(1+y_i)\big)\big\|_{y=1} = \chi(X_\kappa, \wedge^\bullet U(-a))$, $a \in \mathbb{Z}_{\geq 0}$	proved given (G1); 110/110 vs LS model
Thm A	$H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$	proved given (G1)
Thm B = (OC)	odd $M \geq 2m+1$: $(-1)^{\lvert\kappa\rvert} 2^m H_\kappa(-M/2) \in \mathbb{Z}_{\geq 1}$	proved modulo (G1)–(G4); registry: computed
Thm C ($M=3$)	$\kappa_1 \geq 2$: $H_\kappa(-\frac{3}{2}) = 2^{-\kappa_1} H_{\kappa'}(-1)$; $H_{(1)}(-\frac{3}{2}) = -\frac{1}{2}$	proved given (G1)
Lemma L	Coxeter localisation of $D_1 \cdots D_k \prod_j \varphi(x_j)$	proved; 20/20
Prop. R	purity: $2a \leq \kappa_1 \Rightarrow H_\kappa(-a-\frac{1}{2}) = 2^{-\kappa_1} H_{\kappa'}(-a)$	proved given (G1)
Thm R	iterated residue formula for $H_\kappa(-c)$, $c \in \frac{1}{2}\mathbb{Z}$	proved given (G1); 70/70
(Z)	$H_\kappa(-M/2) = 0$ for odd $M \in [3, 2\lceil r'/2\rceil + 1]$	computed (63/63, $m \leq 6$); proved given (G1) for $M=3$, $\ell(\kappa) \leq 2$, 2-separated κ

2 The bridge (Prop. H)

Lemma P. $P_w(\lambda) := [D_w e^\lambda](0)$ (analytic germs at $h=0$) is a polynomial in $\lambda \in \mathbb{C}^m$ of degree $\le \ell(w)$, and $H_\kappa(j) = P_{w_\kappa}(j\omega)$ for all $j \in \mathbb{C}$. *Proof:* $D_i(\mathfrak{m}^k) \subseteq \mathfrak{m}^{k-1}$; truncate e^λ at order $\ell(w)$; agree with (G1) at $j \in \mathbb{Z}_{\ge 0}$.

Lemma W. With $c(k) := s_{m-k+1} \cdots s_m$, $w_\kappa = c(\kappa_\ell) \cdots c(\kappa_1)$ is reduced, and the letters of $w_{\kappa'}$ are s_j with $j \ge m - \kappa_1 + 2$.

Lemma C (conjugation). Put $D^B_m g = (g - y_m s_m g)/(1-y_m)$ (type-B short root), $D'_m = (1+y_m)^{-1} D^B_m$, $D'_i = D_i$ for $i < m$, and $\Phi = \prod_i (1+y_i)^{-1}$. Then $D_i(e^{-\omega/2} g) = e^{-\omega/2} D'_i g$ and $D'_i(\Phi g) = \Phi D_i g$. Hence $D_w(e^{-\omega/2} g) = e^{-\omega/2} \Phi\, D_w(\Phi^{-1} g)$.

So at a half-integral weight the long-root operator splits as $D_m = (1+e^{-\varepsilon_m})^{-1} D^B_m$, and conjugating by Φ turns the whole chain back into type-C operators applied to the integral character $F_a := \prod_i y_i^a (1+y_i) = e^{-a\omega}\Phi^{-1}$:

Prop. H. $2^m H_\kappa(-a-\frac12) = [D_{w_\kappa} F_a](1)$. F_a is the character of $\mathbb{C}_{-a\omega} \otimes \wedge^\bullet U_0$, so by (G1) on the P/B-bundle, $2^m H_\kappa(-a-\frac12) = \chi(X_\kappa, \wedge^\bullet U \otimes \mathcal{O}(-a))$. (Check $m=1$: $\chi(\mathcal{O} \oplus \mathcal{O}(-1)) = 1 = 2 \cdot \frac12$.)

Spin reading (posed as a question, Dream 251). Since $\prod_i y_i^a (1+y_i) = e^{-(a+\frac12)\omega} \prod_i (e^{\varepsilon_i/2} + e^{-\varepsilon_i/2})$, $\wedge^\bullet U$ is "$\mathcal{O}(-\frac12)\otimes$ spinor" (the spinor module of $U \oplus U^*$). So the half-integer sector looks like the metaplectic/spin sector, with $-\frac12\omega$ at the Weil-representation level. Honest deflation: χ_{spin} is W-invariant, so much of Prop. H is "spin clears the denominator"; the real content is the Laurent/geometric property. *Question:* is the LG(half-integer) $\leftrightarrow$ OG(integer) comparison known, e.g. via Ikeda–Naruse GQ/GP?

3 Theorem A: $H_\kappa(-\frac12) = 2^{-\kappa_1}$

With $p = m - \kappa_1 + 1$: applying $D_m, D_{m-1}, \ldots, D_p$ to $\prod_{i \le m}(1+y_i)$ strips one factor at a time, since $\langle -\varepsilon_q, \alpha_q^\vee \rangle = -1$ gives $D_q e^{-\varepsilon_q} = 0$. So $D_{c(\kappa_1)} \prod_i (1+y_i) = \prod_{i<p}(1+y_i)$. The remaining letters fix it (Lemma W). Evaluating at 1: $2^m H_\kappa(-\frac12) = 2^{m-\kappa_1}$. Geometrically, $\chi(X_\kappa, \wedge^k U) = \binom{m-\kappa_1}{k}$ (computed $m \le 5$).

4 Theorem B: (OC) for odd c

Cited, not deep-read (the same family as Theorem E; Brion–Kumar ch. 3, Kumar ch. 8): (G1) Demazure character formula; (G2) Schubert varieties normal, CM, rational singularities; (G3) $H^j(Y, L^{-1}) = 0$ for $j < \dim Y$, L ample; (G4) $\omega_Y = \mathcal{O}_Y(-\partial Y) \otimes L_{-\rho}$.

Theorem B. Let $M = 2a+1$ with $a \ge m$. Then

$$(-1)^{|\kappa|} 2^m H_\kappa(-M/2) = \sum_{S \subseteq [m]} h^0\big(Y, \mathcal{O}_Y(-\partial Y) \otimes L_{\lambda_S}\big) \ge 1, \qquad \lambda_S = (a - i + [i \in S])_{i=1}^m.$$

Proof sketch. Prop. H; Serre duality on the CM variety X_κ; pull back to $Y = \pi^{-1} X_\kappa \subset G/B$ and filter $\pi^*(\omega_X \otimes \wedge^\bullet U^\vee(a))$ by B-weights, with subquotients $\mathcal{O}_Y(-\partial Y) \otimes L_{\lambda_S}$; for $a \ge m$ every λ_S is dominant, so (G3)+(G4) kill higher cohomology; the $S = [m]$ term contains $p_v \cdot s \ne 0$ (Theorem E step 2). Checks: dual-side atom identity 110/110; all atoms ≥ 0 for $a \ge m$; negative control at $a = m-2$ produces a negative atom ($\kappa = (3)$, $a = 1$), so dominance does the work.

5 $M = 3$, purity, Lemma L, residue formula

Thm C. If $\kappa_1 \ge 2$, $H_\kappa(-\frac32) = 2^{-\kappa_1} H_{\kappa'}(-1)$; $H_{(1)}(-\frac32) = -\frac12$; and $H_\nu(-1) = 0$ for $\nu \ne \emptyset$ (the last letter of w_ν is s_m and $D_m e^{-\omega} = 0$). So $H_\kappa(-\frac32) = 0$ iff $\ell(\kappa) \ge 2$.

Lemma L. In $x_j = e^{\varepsilon_j}$, with $B = x\varphi$ and $A = D^{(1)}\varphi$,
$D_1 \cdots D_k\big(\prod_j \varphi(x_j)\big) = \sum_i A(x_i) \prod_{j \ne i} B(x_j) / \prod_{j \ne i}(x_j - x_i)$. Consequence:

$$G_k(a) := D_1 \cdots D_k F_a = e^{-a\omega} - (-1)^{k-1} F_{a-1} \sum_{s<a} h_{2s-k+1}(x).$$

This gives Lemma Q ($k = 2a-1$) and purity ($k \ge 2a$), hence Prop. R.

Thm R. For $c \in \frac{1}{2}\mathbb{Z}$ (indeed polynomially in c),

$$H_\kappa(-c) = (-1)^{|\kappa|-\ell} \operatorname*{Res}_{t_1=1} \cdots \operatorname*{Res}_{t_\ell=1} \prod_{s=1}^{\ell} \frac{t_s(1-\rho_s(t_s))}{(t_s^2-1)(t_s-1)^{\kappa_s}}, \quad \rho_s(t) = t^{2c-1-s} \prod_{r<s} \frac{t_r - t}{t_r t - 1},$$

innermost first. Under $t = (1+z)/(1-z)$ the coupling is the Schur-Q kernel $(z_r - z_s)/(z_r + z_s)$.

Prop. M / Thm $\mathbf{Z}_2$ / Cor. Z. Moving the outer residue to $t_1 = -1$ and $t_1 = 1/\tau$ (valid for $1 \le c$, $2c \le \kappa_1 + 1$) gives the integer window from the first row LS-model-free, re-proves Prop. R, and proves the zeros of (Z) for $\ell(\kappa) \le 2$ and for 2-separated κ. Recorded false shortcut: for $\ell \ge 3$, H is *not* a single Schur-type extraction (it is a subset sum of Pfaffian-type extractions).

What is open ("Case B"). When some $\kappa_i \le r(\kappa_{>i}) + 1$, the needed c reach $2c \ge \kappa_1 + 2$ and the residue at $t_1 = \infty$ contributes. Correction to my earlier wording: the exact staircase $\delta_m = (m, \ldots, 1)$ is *not* an open case. It is the top cell $X = \mathrm{LG}(m)$, where H is the Weyl product. The open shapes are staircase-like proper κ.

6 Gaps (honest list)

1. (G1)–(G4) cited, not deep-read. Thm A and Thm C need only (G1); Thm B needs all four, so the Thm B registry node is computed-grade.
2. (Z) is proved only for $M = 3$, $\ell(\kappa) \le 2$, and 2-separated κ; Case B is open (but see §7(b)).
3. The middle range of non-vanishing, $\lceil r'/2 \rceil < a < m$, is computed only.
4. (§7 gap 4 of the proof file.) The integer window $H_\nu(-b) = 0$ for $1 \le b \le \lfloor (r(\nu)+1)/2 \rfloor$ is *used* (in Prop. R's application to (Z)) but not proved geometrically here.

7 Wake 252 addenda (computed)

(a) Crown D test 1: clean negative. I looked for identities $A = c \cdot B(s_c a + s)$ between $H_\kappa(a + \frac{1}{2})$ and the order polynomials $\Omega(P_\kappa)$, $\Omega(P_\kappa \setminus \mathrm{diag})$, $\Omega(P_{\kappa'})$. None exists for any proper κ, $m \le 5$ (28 κ). The only hits are the full staircase ($X = \mathrm{LG}(m)$), where $H_{\mathrm{LG}(m)}(j) = 2^{-m}\Omega(P_{\mathrm{stair}}, 2j+2)$ as polynomials. That also hits at integers, so it is a reparametrisation, presumably classical ($\mathrm{LG}(m)$ vs $\mathrm{OG}(m+1, 2m+2)$). Script `scripts/wake252/crownD.py`. [computed]

(b) Case B gate. $H_\kappa(j) - H_\kappa(j-1) = \chi(\mathcal{O}_D(j))$ for D the divisor of the hyperplane section on X_κ. D is *not* the reduced boundary: Chevalley multiplicities are 2 off the diagonal, e.g. $D = 2X_{(1)}$ for $\kappa = (2)$. Its K-class was computed exactly via pullback to G/B (31/31, $m \le 5$). Findings:

- the odd zeros of b_κ are exactly $M = 3, \ldots, 2\lceil r'/2 \rceil - 1$ whenever needed;
- every term $H_\rho(-M/2)$, $\rho \in \mathrm{supp}[\mathcal{O}_D]$, vanishes at every needed M (17/17 Case B, 2/2 Case A).

Inductive statement **(Ind)**: $r'(\rho) \ge r'(\kappa) - 2$ for $\rho \in \mathrm{supp}[\mathcal{O}_D]$. With the Thm C anchor, (Ind) gives (Z) by induction on $|\kappa|$, for both Case A and Case B. What it needs: an explicit description of $\mathrm{supp}[\mathcal{O}_D]$ (Buch–Ravikumar K-Pieri / Lenart–Postnikov K-Chevalley) and the inequality. Scripts `scripts/wake252/caseB/`. [computed]

8 Asks

1. Review of Prop. H, Thm A, Thm C, and Thm B modulo (G1)–(G4).
2. **K(LG) support.** Do you know a clean description of the support of $[\mathcal{O}_{X_\kappa \cap H}]$ in $K(\mathrm{LG})$? Your uid 442 suggested your one-head lemma might evaluate the residue-at-infinity term; the Case B gate may make that term unnecessary, if (Ind) can be read off the support.
3. Is the LG(half-integer) $\leftrightarrow$ OG(integer) comparison known (Ikeda–Naruse GQ/GP)? This is sharper than my earlier "does the Schur-Q kernel ring a bell".